

HMD VR Experience Motivates Exercise Intention in Patients with Type 2 Diabetes

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ABSTRACT

Patients are expected to manage their chronic conditions, but very few exercise regularly in addition to taking their medications. Therefore, we developed a 3-minute HMD VR program to educate patients about the importance of exercise. The control group ($n = 30$) received usual care, while the experimental group ($n = 30$) experienced virtual reality and assessed SSQ and exercise intention. No patients experienced VR sickness and the experimental group showed a statistically significant higher intention to exercise for diabetes management. Older adults found the VR experience exciting and novel. For diabetic foot experience, color change is enough to motivate them.

Index Terms: Social and Professional Topics—Computers and Education—Computers in Education—Virtual Reality; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

Managing chronic diseases, including type 2 diabetes, is challenging. In 2020, only one in four patients in South Korea (hereafter Korea) successfully control their blood glucose levels [4]. Patients with uncontrolled diabetes tend to lead to other severe health problems and having diabetic foot. However, patients with diabetes mostly rely on medication management rather than maintaining a healthy lifestyle, including regular exercise, which must be prioritized to control blood glucose levels [5]. According to the Theory of Planned Behavior model [1], having the intention to exercise is important for initiating regular exercise. And the stronger the intention to engage in a behavior, the more likely the behavior should be performed. Patients are the key to chronic disease management, so they should be educated and motivated with the information they need to manage their condition. However, it is difficult to provide patient education in outpatient settings due to staff shortages and lack of environmental support in Korea.

Recently, Virtual reality (VR) has focused on mixed reality engagement for the health and well-being of older adults, and has been applied to combine competition, socialization, and entertaining video material to encourage and facilitate physical activity in older adults [3]. VR games are increasingly being incorporated into

cognitive training for older adults due to their unique immersion and interactivity [2]. The benefits of gamification and the use of VR games in rehabilitation and telerehabilitation have been well reported in post-stroke patients [6]. In addition, HMD VR allows the user to be immersed in a three-dimensional image while wearing an HMD headset that blocks out the external environment. VR can also create a sense of reality and belief in the information obtained through VR if the requirements such as impactful, memorable, and novel are met. In recent years, it has been rapidly developed with lightweight, less fatigue and interactive technologies.

HMD VR has been applied in healthcare for rehabilitation and has proven to be an effective tool for pain management. For the older population, there are concerns that their responses are slow, they may experience more sickness. However, from the feasibility test conducted by our team [7], we realized that older adults are interested in experiencing HMD VR. In response to the educational needs to motivate adults with type 2 diabetes to exercise regularly, we developed an HMD VR patient education program that emphasizes the importance of exercise and diabetic foot prevention. HMD VR is a new way for patients to be educated and motivated while waiting in hospitals or doctors' offices where there is no education room.

2 METHODS

2.1 Implementations

First, we developed the story of educational VR contents about the importance of exercise in diabetes management.

- 1) Through literature review and consultation with experts, including two nurses and two family doctors, we selected the topic of diabetic foot, a multimedia materials that often shown to patients, and outlined the story.
- 2) Meetings were held with VR developers, researchers and consultants to create a final draft and then with experts to validate it.
- 3) We decided the length of the story on the HMD VR about 3 minutes, for the reason that longer stories can lead to loss of concentration and possible dizziness.
- 4) Before implementing VR contents with the story, we validated it with a diabetes specialist and two patients with type 2 diabetes.
- 5) The VR contents with the final version of the storyboard was installed in the HMD.

2.2 Experiments

After IRB approval (HIRB-2023-009), the experimental group was recruited from outpatients in the internal medicine department of a university-affiliated hospital. Inclusion criteria were patients with

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Figure 1: Screenshots of our educational VR contents which motivate diabetes patients to exercise



Figure 2: Experimenting with diabetes patients

type 2 diabetes who agreed to participate in the study and signed a written consent to experience HMD VR. First, we explained the purpose and significance of the study and confirmed that there was no dizziness or other discomforts. We then assessed whether they currently exercised to control their blood glucose levels and asked their intention to exercise on a 10-point Likert scale. After pre-assessment, the patients received usual care from their physician who is a diabetes specialist. And they were educated with a 3-minute HMD VR program developed by research team. After the education, intention to exercise was assessed.

If participants have less fat on their cheeks or, in the case of older people, the skin around their eyes is loose, leaving a gap between the glasses and the nose or cheek, light may pass through and VR may not work accordingly. Therefore, we put on a nose pad for those who has slim cheeks. After recruiting the experimental group, 30 patients with type 2 diabetes, control group recruitment started. They received usual care from the same physician. Their motivation to exercise regularly and what they were currently doing was assessed before and after care.

3 RESULTS

The mean age of the participants was $61.9 (\pm 11.5)$ years and 51.7% were women. They rated their health as fair (5.53 out of 10 score). There were no differences between the groups in terms of age, current level of exercise and intention to exercise. Sixty percent of the participants ($n = 38$, each 16 were allocated in the experimental and control groups) currently engaged in physical activity, with walking being the most prevalent form of exercise. After the experiment, the intention to exercise regularly changed from $7.37 (\pm 2.17)$ to $7.97 (\pm 2.03)$ in the control group and from $7.87 (\pm 2.37)$ to $8.90 (\pm 1.67)$ in the experimental group. ANOVA analysis showed that the experimental group had a statistically significant higher intention

to exercise for diabetes management after education ($F = 3.144$, $P < .012$). At post-assessment, participants were also more likely to select walking as their type of exercise. Regarding the VR experience, no patient in the experimental group experienced VR sickness.

4 IMPLICATIONS

We summarize three implications from this study. First, the study results demonstrated the potential of the HMD VR experience to contribute in patient education including for older adults. The 80% of study participants were over the age of 60, but none of them experienced VR sickness. Rather, they found the VR experience exciting and novel, therefore they were highly engaged during the experience. Second, it was enough to motivate them to exercise by experiencing the process of changing the color of their feet, rather than making them visually uncomfortable by showing them images of diabetic foot. Third, this program is relatively simple, takes short time, is easy to apply to patients, and requires only a small space to run. However, it is important to note that the VR program may be interrupted if the HMD does not fit the older adult's face. Based on the positive results of this study, follow-up research could expand the application of VR experience programs to patients with diabetes for chronic disease management in various clinical settings

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